

From Mesh Avoidance to Catalan Inflation

A proof of the generating-function conjecture for OEIS A289587

A self-contained proof and reproducibility note

September 20, 2026

Abstract

OEIS A289587 counts permutations avoiding the classical pattern 321 and one of two inverse mesh patterns of length two. Its entry attributes an algebraic generating-function conjecture to Thomas Scheuerle, dated December 23, 2025. We prove that formula by identifying exactly what a mesh occurrence means inside a 321-avoiding permutation. Every occurrence is either a pair of fixed points or a pair of entries in an increasing run of consecutive excedance values. Avoidance therefore means at most one fixed point and no upper succession. Contracting maximal upper-succession runs is a reversible operation; at the level of the Narayana generating function it becomes the substitution $u \mapsto u/(1+xu)$. A direct-sum decomposition then gives the conjectured radical expression. We also derive a full mesh-occurrence statistic, multivariate refinements, positive coefficient recurrences and finite sums, and the precise asymptotic $a_n \sim 0.4139310680 \dots (3.5464554447 \dots)^n n^{-3/2}$. The proof is self-contained, and the accompanying exact-arithmetic programs check the literal mesh definition independently of the generating function.

Keywords. Permutation patterns; mesh patterns; Catalan and Narayana numbers; excedances; fixed points; algebraic generating functions.

1 The conjecture and the result

The classical pattern 321 occurs in a permutation $\pi = \pi_1 \cdots \pi_n$ when there are indices $i < j < k$ with $\pi_i > \pi_j > \pi_k$. Write $\text{Av}_n(321)$ for the permutations with no such triple. This is a central Catalan family; the excedance structure and its connection with lattice paths have been studied in detail, for example by Reifegerste [3]. We will prove the particular facts needed here rather than assume them.

A mesh pattern adds empty-region conditions to an ordinary pattern. For the underlying pattern 12, select $i < j$ with $a = \pi_i < b = \pi_j$. The vertical lines through i, j and horizontal lines through a, b divide all other permutation points into nine open rectangles. Label their columns and rows 0, 1, 2, from left to right and from bottom to top. A pair is an occurrence of $(12, R)$ if every rectangle indexed by $(c, r) \in R$ is empty. Points on the selected lines cause no ambiguity, since a permutation has distinct positions and distinct values.

The shading sets relevant to A289587 are

$$R_{174} = \{(0, 1), (0, 2), (1, 0), (1, 2), (2, 1)\}, \quad (1.1)$$

$$R_{234} = \{(0, 1), (1, 0), (1, 2), (2, 0), (2, 1)\}. \quad (1.2)$$

These coordinates agree with the encoding described in the linked SeqFan thread [2]. They also remove any dependence of our proof or programs on an unexplained numerical pattern identifier.

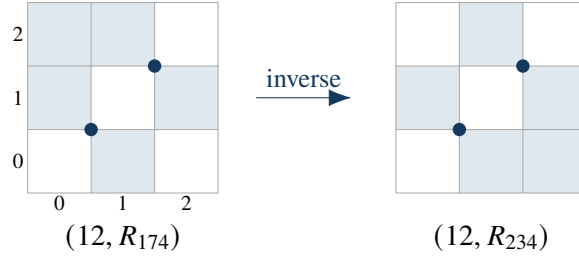


Figure 1: Shaded rectangles must contain no other permutation points. Transposition of the diagram interchanges the two mesh patterns.

Let a_n count $\pi \in \text{Av}_n(321)$ avoiding $(12, R_{174})$, including the empty permutation at $n = 0$. Since $R_{234} = R_{174}^\top$, inversion is a bijection to the class avoiding $(12, R_{234})$. Thus both versions in the OEIS definition have the same counting sequence.

The OEIS entry [1] lists

$$1, 1, 1, 3, 6, 18, 47, 139, 405, 1225, 3740, 11602, \\ 36357, 115049, 366969, 1178791, 3809802.$$

for $n = 0, \dots, 16$, and marks the following formula as conjectural.

Theorem 1.1 (The A289587 generating function). *Set*

$$D(x) = x^4 - 2x^3 - 5x^2 - 2x + 1, \\ P(x) = x^4 + 4x^3 + 11x^2 + 10x + 3, \quad Q(x) = x^2 + 5x + 3.$$

Then

$$A(x) := \sum_{n \geq 0} a_n x^n = \frac{P(x) - Q(x)\sqrt{D(x)}}{8x(1+x)^2}. \quad (1.3)$$

Here $\sqrt{D(x)}$ is the formal square root with constant term 1; the apparent singularity at $x = 0$ is removable.

Status and scope. The entry retrieved for this note still labels (1.3) “Conjectured g.f.” and credits Scheuerle on December 23, 2025. The proof below establishes the exact identity, not merely an initial coefficient agreement. I did not locate a proof of this formula in the entry or its linked discussion. This is a proof of the OEIS-listed conjecture; we make no claim that an exhaustive literature search has ruled out every independent prior derivation. Standard Catalan and Narayana ingredients are explicitly separated from the mesh-specific argument.

2 Elementary structure of 321-avoiders

An index i is an *excedance* of π if $\pi_i > i$, and a *fixed point* if $\pi_i = i$. Let $\text{exc}(\pi)$ and $\text{fix}(\pi)$ denote their numbers. An entry is a *left-to-right maximum* if it exceeds every preceding entry.

Lemma 2.1 (Two increasing subsequences). *In a 321-avoiding permutation, the entries at excedance positions form an increasing subsequence, and the entries at nonexcedance positions form an increasing subsequence. Conversely, these two increasing-subsequence conditions imply 321-avoidance. Every excedance entry is a left-to-right maximum.*

Proof. Suppose $i < j$ are excedances with $\pi_i > \pi_j$. Since $\pi_j > j$, not all values smaller than π_j can occupy the first $j - 1$ positions. Some $k > j$ has $\pi_k < \pi_j$, giving a forbidden triple. The same argument proves the last assertion: if j is an excedance and an earlier entry is larger than π_j , a smaller later value completes a 321.

Now suppose $i < j$ are nonexcedances with $\pi_i > \pi_j$. There are only $\pi_i - 1$ values below π_i , and at least one of them, namely π_j , is later than position i . Because $i \geq \pi_i$, the $i - 1$ preceding positions cannot all contain values below π_i . Some $h < i$ therefore has $\pi_h > \pi_i > \pi_j$, again impossible.

For the converse, any three entries include two belonging to the same one of the two indicated subsequences. Those two entries are increasing, so the three entries cannot be decreasing. $\square$

Lemma 2.2 (A fixed point separates the permutation). *If $\pi \in \text{Av}_n(321)$ and $\pi_i = i$, then all earlier values are less than i and all later values are greater than i .*

Proof. If an earlier value exceeded i , counting the values $1, \dots, i - 1$ would force a later value below i . Together with $\pi_i = i$ these would form a 321. Thus all $i - 1$ preceding values are below i ; they are exactly $1, \dots, i - 1$, and every later value is larger. $\square$

For permutations α of length r and β of length s , their *direct sum* is

$$\alpha \oplus \beta = \alpha_1 \cdots \alpha_r (\beta_1 + r) \cdots (\beta_s + r).$$

A nonempty permutation is sum-indecomposable if it is not a direct sum of two nonempty permutations. Every permutation has a unique decomposition into sum-indecomposable blocks: cut precisely after those $j < n$ for which the first j values form the set $\{1, \dots, j\}$.

By theorem 2.2, fixed points of a 321-avoider are exactly the singleton blocks in this decomposition. No triple 321 straddles a direct-sum boundary, so each block is itself a 321-avoider. These facts will make the fixed-point restriction particularly simple.

3 The exact mesh-occurrence statistic

Definition 3.1. An *upper succession*, also called an upper bond in this note, is a pair of adjacent positions $i, i + 1$ satisfying

$$\pi_i > i, \quad \pi_{i+1} = \pi_i + 1. \quad (3.1)$$

Both positions are then excedances. A maximal upper run is a maximal consecutive block of excedance entries whose values rise by one at each step. Excedance entries not belonging to a longer run count as runs of length one.

Theorem 3.2 (Occurrence formula). *Let $\pi \in \text{Av}_n(321)$ have f fixed points and maximal upper-run lengths $\ell_1, \dots, \ell_k$. Then*

$$\text{occ}_{(12, R_{174})}(\pi) = \binom{f}{2} + \sum_{j=1}^k \binom{\ell_j}{2}. \quad (3.2)$$

More precisely, its mesh occurrences are exactly the pairs of fixed points and the pairs of entries lying in the same maximal upper run.

Proof. For an occurrence with endpoints (i, a) and (j, b) , the shading (1.1) says exactly that

$$\pi_h < a \quad \text{for every } h < i, \quad (3.3)$$

$$a < \pi_h < b \quad \text{for every } i < h < j, \quad (3.4)$$

$$\pi_h \notin (a, b) \quad \text{for every } h > j. \quad (3.5)$$

Every value strictly between a and b must therefore occur strictly between positions i and j , and every entry in those positions has such a value. Consequently

$$j - i = b - a, \quad \{\pi_i, \dots, \pi_j\} = \{a, a + 1, \dots, b\}. \quad (3.6)$$

Also (3.3) forces $a \geq i$.

If $a = i$, then (3.6) gives $b = j$: the endpoints are fixed points. Conversely, any pair of fixed points satisfies (3.3)–(3.5) by theorem 2.2.

If $a > i$, at least one value $c < a$ is missing from the $i - 1$ positions before i . It cannot lie between i and j by (3.4); hence it lies after j . An inversion anywhere in the block from i to j , followed by this value c , would create a 321. The whole block is therefore increasing, and (3.6) makes it

$$a, a + 1, \dots, b.$$

Every one of these positions is an excedance because the value minus the position is the constant $a - i > 0$. Thus the endpoints lie in one upper run.

Conversely, take two entries in one upper run. The first is a left-to-right maximum by theorem 2.1. All intermediate values occur consecutively between the endpoints, so none can occur later. Thus (3.3)–(3.5) hold. The two types of occurrence are disjoint, and counting endpoint pairs proves (3.2). $\square$

Corollary 3.3 (The mesh condition without a mesh diagram). *A permutation $\pi \in \text{Av}_n(321)$ avoids $(12, R_{174})$ if and only if it has no upper succession and has at most one fixed point.*

This is the decisive step: a condition involving five empty rectangles has become one local prohibition and one global restriction on singleton sum blocks.

Example 3.4. The five 321-avoiders of length three are 123, 132, 213, 231, 312. The first has three fixed points, while 231 has the upper succession 2, 3. The remaining three are accepted, giving $a_3 = 3$. The permutation 1324 has no upper succession, but its two fixed points, 1 and 4, form a mesh occurrence. Thus the fixed-point condition cannot be omitted.

4 Contracting upper runs

The next step is an exact decomposition, not an inclusion–exclusion approximation. Contract every maximal upper run to one entry and standardize the resulting word, meaning that its smallest remaining value becomes 1, the next becomes 2, and so on. Call the result the *upper-run core*.

Theorem 4.1 (Canonical core and inflation). *Every 321-avoiding permutation has a unique representation as a 321-avoiding core with no upper succession, with each excedance entry of the core inflated into a nonempty increasing interval. The lengths of the inflated intervals are precisely the original maximal upper-run lengths. This operation preserves the number of fixed points and preserves the direct-sum block structure.*

Proof. We first justify a single contraction. Suppose adjacent positions $i, i + 1$ contain $a, a + 1$ with $a > i$. Delete the second entry and standardize. Pattern avoidance survives taking a subpermutation. All entries before i are below a by theorem 2.1, so they are unchanged. The surviving entry a remains an excedance.

For an entry later than $i + 1$, if its value is larger than $a + 1$, both its position and value decrease by one; its excedance or fixed-point status is unchanged. Suppose instead that this later entry is $b < a$ at position j . Every value smaller than b must precede j : otherwise a, b , and a later smaller value would form a 321. The $j - 1$ preceding positions contain those $b - 1$ smaller values and also both $a, a + 1$. Therefore $j - 1 \geq b + 1$, or $b \leq j - 2$. After contraction, $b < j - 1$, so it remains a strict nonexcedance. Thus precisely one excedance disappears, and no fixed point appears or disappears.

It remains important to check that contraction creates no unexpected new upper bond. If two consecutive excedance entries have a gap between their values, every value in that gap occurs later: the first entry is a left-to-right maximum and there is no position between the two. Such intervening values cannot be later excedances, since excedance values increase; they are nonexcedances and are never contracted. Their presence therefore prevents the gap from disappearing. Likewise, a nonexcedance between two upper runs is never deleted, so separated runs cannot become adjacent. Contracting a bond within a run merely shortens that run. Hence the maximal runs have a uniquely determined bond-free core, independent of contraction order.

Conversely, inflate any excedance of a bond-free core into a consecutive increasing block. Increasing interval inflation preserves 321-avoidance: a decreasing triple uses at most one entry from such a block, and would project to a decreasing triple in the core. Entries before the inflated point have smaller values and are unchanged; later larger entries have both position and value shifted by the same amount, while later smaller entries stay strict nonexcedances. Fixed points remain fixed, and all entries of the inflated block are excedances.

A pre-existing position gap between core excedances contains a retained nonexcedance; a pre-existing value gap contains a retained nonexcedance value, by the preceding argument. Hence different inflated blocks never merge into one upper run. The inflation lengths and the core are therefore recovered uniquely by contraction.

Finally, an upper bond cannot cross a direct-sum boundary. If it did, its first value, being a left-to-right maximum and lying in the left summand, would have to equal the boundary position, contradicting strict excedance. Inflating an entry of a nontrivial indecomposable block preserves indecomposability: a sum cut between distinct inflated blocks would project to a sum cut in the core; a cut inside an inflated block is impossible because its first entry still exceeds its position and leaves a smaller value on the right. Thus the sum blocks correspond under contraction and inflation. $\square$

Example 4.2. The permutation

$$2341675 = 2341 \oplus 231$$

has upper runs of lengths 3 and 2. Its core is $2143 = 21 \oplus 21$. Inflating the two excedances of 2143 by those lengths reconstructs 2341675. The occurrence formula gives $\binom{3}{2} + \binom{2}{2} = 4$ mesh occurrences.

5 The Narayana generating function, from first principles

Define

$$C(x, u) = \sum_{n \geq 0} \sum_{\pi \in \text{Av}_n(321)} x^n u^{\text{exc}(\pi)}. \quad (5.1)$$

We need its functional equation with u still present; substituting the ordinary Catalan series $C(x, 1)$ too early would lose exactly the information needed for upper-run contraction.

Lemma 5.1 (Excedance arrays and Dyck valleys). *Permutations in $\text{Av}_n(321)$ with k excedances are in bijection with Dyck paths of semilength n having k valleys.*

Proof. Let $e_1 < \dots < e_k$ be the excedance positions and $v_1 < \dots < v_k$ their values. By theorem 2.1, the remaining values occupy the remaining positions in increasing order. The pair of arrays therefore determines the permutation, and satisfies $e_j < v_j$.

Conversely, choose increasing arrays in $\{1, \dots, n\}$ with $e_j < v_j$, put v_j in position e_j , and fill the remaining positions increasingly. For every t , the number of selected positions at most t is at least the number of selected values at most t , since $v_j \leq t$ implies $e_j < t$. Thus the number of unselected values at most t is at least the number of unselected positions at most t . Increasing matching places each remaining value at a position at least as large as itself. These are precisely the nonexcedances, and the two increasing subsequences imply 321-avoidance.

Use north and east steps for a Dyck path from $(0, 0)$ to (n, n) staying in $y \geq x$. Specify its valleys (an east step followed by a north step) to be

$$(e_1, v_1 - 1), \dots, (e_k, v_k - 1).$$

Both coordinates increase strictly, and $e_j \leq v_j - 1$. These valleys uniquely determine a path: alternate maximal north and east runs, starting with a north run and ending with an east run. Each valley is on or above the diagonal, so the whole path stays above it. Conversely, the valleys of such a path recover exactly the allowed arrays. The path with no valleys is $N^n E^n$, corresponding to the identity permutation. $\square$

Proposition 5.2 (Narayana equation). *The series $C(x, u)$ is the unique formal solution with constant term 1 of*

$$C = 1 + xC(1 - u + uC). \quad (5.2)$$

Equivalently, for $n \geq 1$,

$$[x^n][u^k]C(x, u) = \frac{1}{n} \binom{n}{k} \binom{n}{k+1}, \quad 0 \leq k < n. \quad (5.3)$$

Proof. A nonempty Dyck path decomposes uniquely as $NPEQ$, where P, Q are Dyck paths. There is a new valley between the indicated E and the start of Q exactly when Q is nonempty. Thus the generating function is $1 + xC(1 + u(C - 1))$, proving (5.2). The factor x determines each coefficient from earlier ones, giving uniqueness.

For the coefficient formula put $Y = C - 1$. Then $Y = x(1 + Y)(1 + uY)$. Formal Lagrange inversion gives

$$[x^n]Y = \frac{1}{n} [z^{n-1}](1 + z)^n (1 + uz)^n.$$

Extracting u^k proves (5.3). This recovers the classical Narayana distribution of excedances (cf. [3, Corollary 3.7]) with the indexing conventions used here. $\square$

6 From contraction to the conjectured radical

Let $B(x, u)$ count all upper-bond-free 321-avoiders by length and excedance number, with any number of fixed points. A core excedance has weight xu . Inflating it into a positive run changes this weight to

$$xu + (xu)^2 + (xu)^3 + \dots = \frac{xu}{1 - xu}.$$

The canonical decomposition in theorem 4.1 gives

$$C(x, u) = B\left(x, \frac{u}{1 - xu}\right), \quad \boxed{B(x, u) = C\left(x, \frac{u}{1 + xu}\right)}. \quad (6.1)$$

The two substitutions are compositional inverses as formal power series in x . No analytic convergence assertion is involved.

At $u = 1$, write simply $B(x) = B(x, 1)$. Substitution into (5.2) yields

$$xB^2 - (1 + x - x^2)B + 1 + x = 0, \quad B(x) = \frac{1 + x - x^2 - \sqrt{D(x)}}{2x}. \quad (6.2)$$

The other algebraic branch has a pole at zero, whereas this branch has constant term 1.

Let $I(x)$ count the nontrivial sum-indecomposable blocks in this bond-free class. Upper bonds do not cross sum boundaries, so arbitrary bond-free permutations are sequences of singleton blocks, of weight x , and nontrivial blocks, counted by I . Therefore

$$B = \frac{1}{1 - x - I}, \quad I = 1 - x - \frac{1}{B} = \frac{1 - x - x^2 - \sqrt{D(x)}}{2(1 + x)}. \quad (6.3)$$

The empty permutation is included in B , but not in I .

Let $H(x)$ count the bond-free permutations with no fixed points. They are sequences of nontrivial blocks, so

$$H = \frac{1}{1 - I} = \frac{B}{1 + xB} = \frac{1 + 3x + x^2 - \sqrt{D(x)}}{4x(1 + x)}. \quad (6.4)$$

By theorem 3.3, an accepted permutation either has no fixed point, or has one fixed point with an arbitrary fixed-point-free bond-free permutation on either side. It follows that

$$\boxed{A(x) = H(x) + xH(x)^2}. \quad (6.5)$$

This identity has a literal bijective interpretation: split at the unique singleton block when one exists.

Completion of the proof of theorem 1.1. Substitute (6.4) into (6.5) and replace $(\sqrt{D})^2$ by D . The rational part of the resulting numerator is $P(x)$ and the coefficient of $\sqrt{D}$ is $-Q(x)$, with denominator $8x(1 + x)^2$, exactly as in (1.3). Every prior identity was derived from a bijection or a unique formal functional equation, so this proves the formula for all coefficients simultaneously. $\square$

Why the quartic occurs. The discriminant of the Narayana equation becomes

$$(1 + x - x^2)^2 - 4x(1 + x) = D(x)$$

after the upper-run substitution. Thus the unusual quartic is not an accident of guessing from a short initial segment: it is the ordinary Catalan discriminant transformed by a canonical run contraction.

7 Refinements beyond avoidance

The same proof retains substantially more information than a single univariate generating function. Define

$$H(x, u) = \frac{B(x, u)}{1 + xB(x, u)}, \quad B(x, u) = C\left(x, \frac{u}{1 + xu}\right). \quad (7.1)$$

This counts the upper-bond-free 321-avoiders with no fixed points, by length and excedances.

Theorem 7.1 (Fixed-point refinement). *The generating function for upper-bond-free 321-avoiders, with t marking fixed points, is*

$$\mathcal{B}(x, u, t) = \frac{H(x, u)}{1 - txH(x, u)}. \quad (7.2)$$

In particular, the subfamily with exactly r fixed points has generating function

$$x^r H(x, u)^{r+1}. \quad (7.3)$$

The excedance-refined generating function for A289587 is $H(x, u) + xH(x, u)^2$.

Proof. Split a permutation at its r singleton sum blocks. This gives an ordered $(r+1)$ -tuple of fixed-point-free pieces, including empty pieces, and r singleton blocks. The decomposition is unique, and excedance number is additive under direct sum. Summing over r gives (7.2). $\square$

One can also enumerate the number of mesh occurrences itself. Let

$$W(x, u, q) = \sum_{\ell \geq 1} x^{\ell-1} u^\ell q^{\binom{\ell}{2}} = u + xu^2q + x^2u^3q^3 + x^3u^4q^6 + \cdots. \quad (7.4)$$

Multiplying W by x gives the weight of one inflated upper run.

Theorem 7.2 (All mesh occurrences). *For all 321-avoiding permutations, the joint generating function of length, excedances, fixed points, and occurrences of $(12, R_{174})$ is*

$$\begin{aligned} \mathcal{F}(x, u, t, q) &:= \sum_{n \geq 0} \sum_{\pi \in \text{Av}_n(321)} x^n u^{\text{exc}(\pi)} t^{\text{fix}(\pi)} q^{\text{occ}_{(12, R_{174})}(\pi)} \\ &= \sum_{r \geq 0} (tx)^r q^{\binom{r}{2}} H(x, W(x, u, q))^{r+1}. \end{aligned} \quad (7.5)$$

Proof. In a fixed-point-free core, replace the weight xu of each excedance by $xW(x, u, q)$. The inflated run contributes $\binom{\ell}{2}$ occurrences by theorem 3.2. Insert r fixed points as in theorem 7.1; together they contribute $\binom{r}{2}$ occurrences, independently of all upper runs. No other occurrences exist. This gives (7.5). $\square$

Every coefficient of x in (7.5) is a polynomial in the other variables, so all substitutions are formally well-defined. At $q = 0$, only run length one and $r = 0, 1$ survive; this recovers the avoidance formula. At $q = 1$, $W = u/(1 - xu)$, undoing core contraction, and at $t = 1$ the result reduces to $C(x, u)$.

Corollary 7.3 (Counting upper bonds rather than mesh occurrences). *If s marks the number of adjacent upper bonds, then the joint generating function for all 321-avoiders is*

$$\frac{C\left(x, \frac{u}{1 + (1-s)xu}\right)}{1 + (1-t)x C\left(x, \frac{u}{1 + (1-s)xu}\right)}. \quad (7.6)$$

Proof. A run of length ℓ has $\ell - 1$ adjacent upper bonds, so its series is $xu/(1 - sxu)$. Apply (6.1) and then (7.2). $\square$

The difference between (7.5) and (7.6) is essential: a length-three run gives two adjacent bonds but three mesh occurrences.

8 Exact coefficients: recurrences and finite sums

8.1 A cancellation-free quadratic recurrence

The radical expression for I is equivalent to

$$\boxed{I = x^2 + (x + x^2)I + (1 + x)I^2, \quad I(0) = 0.} \quad (8.1)$$

Its unique solution starts at degree two. Write $i_n = [x^n]I$ and $h_n = [x^n]H$, and take negatively indexed coefficients to be zero. Then $i_0 = i_1 = 0$, $h_0 = 1$, and, for $n \geq 2$,

$$i_n = \mathbf{1}_{n=2} + i_{n-1} + i_{n-2} + \sum_{j=0}^n i_j i_{n-j} + \sum_{j=0}^{n-1} i_j i_{n-1-j}. \quad (8.2)$$

The apparent terms involving i_n vanish because $i_0 = i_1 = 0$. For $n \geq 1$,

$$h_n = \sum_{j=2}^n i_j h_{n-j}, \quad (8.3)$$

$$a_n = h_n + \sum_{j=0}^{n-1} h_j h_{n-1-j}, \quad a_0 = 1. \quad (8.4)$$

These formulas use only addition and multiplication of nonnegative integers. They compute the first N terms in $O(N^2)$ arithmetic operations and $O(N)$ stored integers; this is an arithmetic-operation bound, not a claim about bit complexity for growing integer sizes.

Table 1: The building-block sequences and the first terms of A289587.

n	i_n	h_n	$[x^n]B$	a_n
0	0	1	1	1
1	0	0	1	1
2	1	1	2	1
3	1	1	4	3
4	3	4	10	6
5	7	9	26	18
6	19	27	72	47
7	53	76	206	139
8	153	227	606	405
9	453	683	1820	1225
10	1367	2096	5558	3740

8.2 A minimal polynomial for the target series

Squaring the radical formula and simplifying gives

$$\boxed{4x(1+x)^2 A^2 - P(x)A + (1+x)(2x^2 + 6x + 3) = 0.} \quad (8.5)$$

The coefficient of A at $x = 0$ is -3 , so the branch $A(0) = 1$ is uniquely determined as a formal series. This provides another exact certificate of the formula. Since D has distinct roots and Q is not zero, $\sqrt{D}$ is not rational and neither is A ; the algebraic degree over $\mathbb{Q}(x)$ is exactly two. In particular, the counting sequence does not satisfy any eventual constant-coefficient linear recurrence.

8.3 A positive finite-sum formula

Let $\text{Cat}_j = (j+1)^{-1} \binom{2j}{j}$. Solving (8.1) in Catalan form yields

$$I(x) = \sum_{j \geq 0} \text{Cat}_j \frac{x^{2j+2}(1+x)^j}{(1-x-x^2)^{2j+1}}. \quad (8.6)$$

This follows either by substituting the Catalan equation or by factoring the discriminant as

$$D = (1-x-x^2)^2 \left(1 - \frac{4x^2(1+x)}{(1-x-x^2)^2} \right).$$

Expanding the two remaining factors gives, for $n \geq 2$,

$$i_n = \sum_{j=0}^{\lfloor (n-2)/2 \rfloor} \text{Cat}_j \sum_{\ell=0}^j \binom{j}{\ell} \sum_{\substack{a,b \geq 0 \\ a+2b=n-2j-2-\ell}} \frac{(2j+a+b)!}{(2j)! a! b!}. \quad (8.7)$$

Indeed, the coefficient of x^{a+2b} in $(1-x-x^2)^{-m}$ contributed by a copies of x and b copies of x^2 is $(m+a+b-1)!/((m-1)!a!b!)$.

For a completely recurrence-free formula for a_n , let $C_{\geq 2}(m, k)$ be the ordered k -tuples of integers at least two whose sum is m . For $k = 0$ it consists of the empty tuple if $m = 0$ and is empty otherwise. Then

$$\begin{aligned} a_n = & \sum_{k=0}^{\lfloor n/2 \rfloor} \sum_{(m_1, \dots, m_k) \in C_{\geq 2}(n, k)} \prod_{j=1}^k i_{m_j} \\ & + \sum_{k=0}^{\lfloor (n-1)/2 \rfloor} (k+1) \sum_{(m_1, \dots, m_k) \in C_{\geq 2}(n-1, k)} \prod_{j=1}^k i_{m_j}. \end{aligned} \quad (8.8)$$

The second outer sum is empty at $n = 0$. Together with (8.7), this is a finite formula involving only binomial coefficients, factorials, and nonnegative sums and products. It follows directly from

$$A = \sum_{k \geq 0} I^k + x \sum_{k \geq 0} (k+1) I^k.$$

The factor $k+1$ is also the number of places to insert one singleton among k nontrivial blocks.

8.4 An independent alternating coefficient formula

The Narayana substitution provides a different computational route: for $n \geq 1$,

$$[x^n]B(x) = 1 + \sum_{m=2}^n \sum_{k=1}^{m-1} (-1)^{n-m} \frac{1}{m} \binom{m}{k} \binom{m}{k+1} \binom{k+n-m-1}{n-m}. \quad (8.9)$$

Here the initial 1 accounts for zero excedances. This follows by expanding $(1+x)^{-k}$ in $B(x) = C(x, (1+x)^{-1})$. Unlike (8.2)–(8.4), it has cancellation, but precisely for that reason it is useful as an independent exact-arithmetic check.

9 Asymptotic enumeration

The radical formula also determines the exponential growth and its polynomial correction. Factor

$$D(x) = (x^2 - (1 + 2\sqrt{2})x + 1)(x^2 + (2\sqrt{2} - 1)x + 1). \quad (9.1)$$

The first factor has two positive reciprocal roots, and the second has two nonreal roots on the unit circle. Hence the smallest-modulus zero is the positive number

$$\rho = \frac{1 + 2\sqrt{2} - \sqrt{5 + 4\sqrt{2}}}{2} = 0.2819716800611948531 \dots \quad (9.2)$$

The rational denominator in (1.3) has no nonremovable zero inside the unit disk. Its numerator has a nonzero square-root coefficient at ρ , so ρ is the unique dominant singularity.

Theorem 9.1 (Precise leading asymptotic). *Let*

$$\gamma = \rho^{-1} = \frac{1 + 2\sqrt{2} + \sqrt{5 + 4\sqrt{2}}}{2} = 3.5464554446849952445 \dots, \quad (9.3)$$

$$\delta = \sqrt{-\rho D'(\rho)}, \quad K = \frac{(\rho^2 + 5\rho + 3)\delta}{16\sqrt{\pi}\rho(1 + \rho)^2} = 0.4139310680369706448 \dots \quad (9.4)$$

Then

$$a_n \sim K \gamma^n n^{-3/2}. \quad (9.5)$$

Proof. Put $z = 1 - x/\rho$. The root is simple and $D'(\rho) < 0$, so

$$\sqrt{D(x)} = \delta z^{1/2}(1 + O(z)).$$

Write $A = R - S\sqrt{D}$, where

$$R(x) = \frac{P(x)}{8x(1+x)^2}, \quad S(x) = \frac{Q(x)}{8x(1+x)^2}.$$

The leading nonanalytic term is $-S(\rho)\delta z^{1/2}$. The binomial-coefficient asymptotic

$$[x^n](1 - x/\rho)^{1/2} = -\frac{\rho^{-n}}{2\sqrt{\pi}n^{3/2}}(1 + O(n^{-1}))$$

follows from the gamma-function expression for $(-1)^n \binom{1/2}{n}$ and Stirling's formula. A slit neighborhood extends to a disk of radius strictly larger than ρ without meeting another singularity. The convergent local square-root expansion therefore transfers termwise; the next nonanalytic term is of order $z^{3/2}$ and contributes $O(\rho^{-n}n^{-5/2})$. Equivalently, this is the usual square-root singularity transfer, obtained by deforming a Cauchy coefficient contour to the slit at ρ . Thus the leading constant is $S(\rho)\delta/(2\sqrt{\pi})$, namely (9.4). $\square$

The first relative correction is also explicit. Define

$$c_1 = \frac{3}{8} - \frac{3}{2} \left(\frac{\rho^2 D''(\rho)}{4\delta^2} - \frac{\rho S'(\rho)}{S(\rho)} \right) = -1.0082530378112395475 \dots \quad (9.6)$$

Expanding one term further gives

$$a_n = K \gamma^n n^{-3/2} \left(1 + \frac{c_1}{n} + O(n^{-2}) \right). \quad (9.7)$$

For completeness, the coefficient of z in $S(\rho(1-z))\sqrt{D(\rho(1-z))}/(S(\rho)\delta\sqrt{z})$ is the parenthesized expression in (9.6). The correction $3/8$ comes from the coefficient asymptotic for $z^{1/2}$, while the factor $-3/2$ is $\Gamma(-1/2)/\Gamma(-3/2)$.

Table 2: Exact coefficients divided by the leading asymptotic.

n	$a_n/(K\gamma^n n^{-3/2})$
20	0.951983446713366...
50	0.980230192459601...
100	0.990017308750539...
200	0.994983824600422...

Corollary 9.2 (A limiting fixed-point probability). *Choose an A289587 permutation uniformly at random from those of length n . The probability that it has exactly one fixed point tends to*

$$\frac{2\rho H(\rho)}{1 + 2\rho H(\rho)} = 0.4288852556670782555\dots, \quad H(\rho) = \frac{1 + 3\rho + \rho^2}{4\rho(1 + \rho)}. \quad (9.8)$$

Proof. Write $H = h_0 - h_1\sqrt{1 - x/\rho} + O(1 - x/\rho)$, where $h_1 > 0$. The corresponding square-root coefficients in $H + xH^2$ and xH^2 are $-h_1(1 + 2\rho h_0)$ and $-2\rho h_0 h_1$. Their coefficient ratio gives (9.8). $\square$

10 Reproducibility and proof audit

The archive contains a standard-library Python verifier, `code/verify.py`, and an optional SymPy/mpmath script, `code/symbolic_checks.py`. Neither program downloads data or uses numerical guessing to establish the formula.

Independent enumeration. To generate all 321-avoiders, insert a new maximum into every position whose following suffix is increasing. A newly created 321 would have to start at the new maximum, so this criterion is necessary and sufficient. Every resulting permutation has a unique parent obtained by deleting its maximum. Catalan totals are checked at each length. For mesh membership, the program tests the nine-rectangle definition directly, rather than replacing it by theorem 3.3. This is important because the structural theorem is exactly the result that the independent test should be capable of falsifying.

Table 3: Executed checks included in the reproducibility logs.

Check	Range
Literal mesh test, both R_{174} and R_{234} , against the structural characterization and coefficient recurrence	$0 \leq n \leq 12$
Structural counts and every fixed-point refinement against the coefficient recurrences	$0 \leq n \leq 13$
Core contraction/inflation round trips and exact occurrence identity (3.2)	$0 \leq n \leq 10$
Radical expansion versus the positive recurrences	$0 \leq n \leq 200$
Narayana-substitution finite sum (8.9)	$0 \leq n \leq 60$
Positive finite sum (8.7)	$0 \leq n \leq 45$
Published OEIS initial values	$0 \leq n \leq 16$

All checks passed. At length 12, the literal test examines all 208012 Catalan permutations and counts 36357 accepted ones; at length 13, the structural test examines all 742900 Catalan permutations and counts 115049. The symbolic script verifies seven algebraic residuals identically equal to zero, including (6.5), (8.5),

and (9.1). The coefficient file extends the sequence through $n = 200$ and is explicitly identified as computed from this proof, not as an official OEIS file.

What constitutes the proof. Finite tests are corroboration, not a substitute for the argument. The proof consists of four logically separate steps: the exact mesh occurrence classification; reversible upper-run contraction; the Narayana functional equation with an excedance marker; and the fixed-point/direct-sum decomposition. All later identities are algebraic consequences of those steps. There is no extrapolation from a guessed recurrence, no unverified infinite-state truncation, and no dependence on checking only the currently displayed OEIS terms.

Rebuilding. From the archive directory, run

```
python3 code/verify.py
python3 code/symbolic_checks.py          # optional; needs sympy, mpmath
pdflatex -interaction=nonstopmode article.tex
pdflatex -interaction=nonstopmode article.tex
```

The first program writes coefficient tables and a machine-readable verification report to `data/`. The second writes exact symbolic check results and high-precision numerical approximations. This note provides a conventional mathematical proof; it is not a proof-assistant formalization, and its numerical approximations are not presented as interval-arithmetic certificates.

11 Conclusion

The generating-function conjecture for A289587 is true. The proof rests on an exact classification of mesh occurrences as fixed-point pairs or pairs in one upper run. Reversible run contraction gives $B(x, u) = C(x, u/(1 + xu))$, and the direct-sum decomposition gives $H = B/(1 + xB)$ and $A = H + xH^2$. Thus the conjectured quartic radical comes from the Narayana equation after a canonical change of variables, not from extrapolating a finite coefficient list. The same structure yields refined occurrence enumeration, positive recurrences and finite formulas, and a precise asymptotic expansion.

References

- [1] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A289587, <https://oeis.org/A289587>. Sequence initiated by N. J. A. Sloane; generating-function conjecture attributed in the entry to Thomas Scheuerle, December 23, 2025. Entry consulted September 20, 2026.
- [2] Christian Sievers and other contributors, *RFE Dec 2025: Mesh patterns avoiding 321*, SeqFan discussion, December 2025, <https://groups.google.com/g/seqfan/c/9jqIgQGj8Gg>. The thread specifies the mesh-shading encoding and identifies the pair of patterns numbered 174 and 234. Consulted September 20, 2026.
- [3] Astrid Reifegerste, *The excedances and descents of bi-increasing permutations*, preprint, 2002, arXiv:math/0212247, <https://arxiv.org/abs/math/0212247>. In particular, the bi-increasing characterization, the fixed-point separation property, and Corollary 3.7 on the Narayana distribution provide background for the self-contained preliminaries above.